Sampling and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Sampling and Representation

> Prepare Sampling and Representation: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Define a question and target **population**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Missing Data

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Look for: Detect absent and unusable values

Reference example, not the only acceptable wording: 0 is a recorded allowed duration. The other two are unknown under this schema.

Overfitting Intuition

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

Look for: Build a labelled confusion matrix from matched reference and **prediction** pairs

Reference example, not the only acceptable wording: Reference clear/predicted clear contains V1,V2: 2. Clear/blocked contains V3: 1. Blocked/clear contains V4,V5: 2. Blocked/blocked contains V6,V7,V8: 3. Matrix [[2,1],[2,3]].

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 **response**s may still fail to represent the school. Propose a better sampling plan using fictional data.

Concept to explain

Week 19: Sampling and **Representation**

Term	Meaning
sample	A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.
population	The full group of cases or people a study aims to describe.
representation	A way to encode information so a person or system can work with it.
sampling bias	A systematic mismatch between selected examples and the population the study intends to describe.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Teach Week 19 as a two-session investigation using fictional data. A fictional school wants to know which activity Grade 7 students prefer.

Surveying only the coding club is easy, but the question concerns every Grade 7 student. First define the **population**, selection **process**, question and **response** options. No real student data is required for this exercise.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

6/10 = 60 percent among those **response**s.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Apply the connected idea: Graphs and honest scales

Teach Week 18 as a two-session investigation using fictional data. A fictional poll records 8 votes for Drawing and 10 for Building. A **bar chart** should make lengths proportional to counts, starting at zero.

A chart starting at 7 makes the visible bars 1 and 3 units tall and can exaggerate the difference. Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

2.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

End goal for the pictorial case

By the end, I can define a **population**, selection frame and non**response** report for a survey.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

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Explain the mechanism

> Model Sampling and Representation using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Goal: books for the whole school.

2. **Sample**: one advanced-reading club at lunchtime.

3. Missing: younger readers, other interests, absent students, different access needs.

4. Repair collection across grades, times, interests, and access needs; then recheck.

Recalculate, retrace, or restate the reasoning in your own **representation**:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

Only adventure-club members answer a school reading survey.

The teaching case

Explain the sampling problem.

Exact explanation to model

Why the result follows

Membership selects for a relevant interest. Include relevant grades and interests with

appropriate data practices.

Explain the added worked example

Explain the added worked example · 1

A fictional school wants to know which activity Grade 7 students prefer. Surveying only the coding club is easy, but the question concerns every Grade 7 student.

First define the **population**, selection **process**, question and **response** options. No real student data is required for this exercise.

Explain the added worked example · 2

The result 2 of 3 describes these fictional club **response**s. It cannot by itself estimate the whole grade reliably. A better plan starts from a complete grade roster, uses a fair selection method and records non**response**.

Random selection gives each listed student a chance; it does not guarantee every small **sample** is representative.

Key terms

Term	Meaning
Population	The full group a question concerns
Sample	The observed subset of that group
Sampling frame	The list or source from which a sample is selected
Response	A recorded answer from a selected participant

Apply the connected idea: Graphs and honest scales

A fictional poll records 8 votes for Drawing and 10 for Building. A **bar chart** should make lengths proportional to counts, starting at zero.

A chart starting at 7 makes the visible bars 1 and 3 units tall and can exaggerate the difference.

The text chart uses one # per vote and a zero **baseline**. The second category has 2 more votes and 1.25 times as many votes, not three times as many.

Include the poll question, date or collection occasion and number of **response**s when presenting a real graph.

Difference 2 votes; ratio 6/4 = 1.5. Both bars begin at zero. Exit: No. Preference counts do not measure learning effectiveness or establish a cause.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

> **If the school adds 20 more club responses, which coverage problem remains?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sampling and Representation

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

The facts we will use

Supplied facts

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Supplied facts

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

AI ACADEMY / AI-2/19

Sampling and Representation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Define target

The question concerns all Grade 7, not only library-club members.



2. Select invitations

Teacher uses the complete approved list and documented random selection.



3. Record responses

12 replies from 15 invited; three nonresponses remain visible.



4. Report scope

Invitation response rate 12/15=80%; preference counts describe respondents.

CHANGE ONE THING

Invite only the 15 library-club members instead.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define target	The question concerns all Grade 7, not only library-club members.
2. Select invitations	Teacher uses the complete approved list and documented random selection.
3. Record responses	12 replies from 15 invited; three nonresponses remain visible.
4. Report scope	Invitation response rate 12/15=80%; preference counts describe respondents.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

What this example does not establish

Keep the stated limit

Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

Change one thing in the pictured case

Change one thing

Change: Invite only the 15 library-club members instead.

Predicted result

Prediction: The invitation **process** no longer covers the whole grade through this frame.

Why the result follows

Reason: Club members may differ in reading interests; selection is now limited to one group.

Ask while showing the picture

Ask while showing the picture · 3

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a **representation**; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Sampling and Representation, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: Only adventure-club members answer a school reading survey.

2. State the decision to explain

Write the question to explain: Explain the sampling problem.

3. Reveal the worked reasoning

Reveal after **prediction**s: Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Hold back the independent task

Keep for an independent attempt: You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Students outside the club are still not represented by the selection **process**. Exit: No. The 12 nonrespondents may differ. Report 8 of 20 and the limitation.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Completion practice

Completion task: Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

Reasoning to listen for

Teacher check: 7/12 of these respondents prefer longer lunch, about 58.3 percent.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a **response** from every learner before discussion.

Individual reasoning check

Use the worked case for Questions **population**s and **sample**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> #### Expected repair
>
> Expected correction: Selection and **response** are separate. Even a fair invitation **sample** can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Connected teaching cycle: Graphs and honest scales

Target: Choose an appropriate graph

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Counts are 12 and 15. State the difference and ratio using the true values.

> #### Reasoning to listen for
>
> Teacher check: Difference 3; ratio 15/12 = 1.25.

Individual check: For Graphs and honest **scale**s, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected repair

Expected correction: Counts alone do not show why students chose an activity or whether nonrespondents would choose differently.

Graph choices should match the question: categories suit bars; ordered time **observation**s can suit a line.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

Pictorial practice question

Can **80% response** be reported as **80%** of the whole grade responding?

Reasoned teacher answer

No. Twelve of fifteen invited is **80%**; twelve of sixty in the grade is 20%. Denominators answer different questions.

Changed-case reasoning: The invitation **process** no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the weekly evidence lab

> Run the Sampling and Representation investigation and use the first unsupported step to choose a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Compare **sample** cards with an intended-user checklist, highlight missing settings and users, and design a broader collection plan.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

A successful run proves the conclusion: Selection and **response** are separate. Even a fair invitation **sample** can be distorted if only interested students respond.

Report numbers invited and responding and explain who may be absent.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: plan a **sample** about library opening times for the whole grade.

A library-club-only survey fails coverage; require a full-grade frame and non**response** plan.

A convenient shortcut preserves meaning: Changing the report heading from club **response**s to all students does not fix sampling. Change the collection plan or narrow the claim.

Avoid collecting names when anonymous counts are sufficient; a teacher can manage selection without publishing identities.

Check these misconceptions

A successful run proves the conclusion: Counts alone do not show why students chose an activity or whether nonrespondents would choose differently.

Graph choices should match the question: categories suit bars; ordered time **observation**s can suit a line.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: Reading 4, Building 8 and Drawing 6; draw all bars from zero with one **scale**.

Building has twice Reading, and 2 more than Drawing.

A convenient shortcut preserves meaning: A graph with bars beginning at 7 turns counts 8 and 10 into visible lengths 1 and 3. Redraw from zero and label counts directly.

A line chart can use a restricted **axis** for detail if clearly labelled, but it must not disguise its **scale**.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Sampling and Representation questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

> **Teacher key · Worked case answer**
>
> Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Independent question

Independent question · withhold the key

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

Expected independent answer

Teacher key · Expected independent answer

Chess-club readers may not represent all students. Include learners from other groups and reading interests; broader coverage still needs honest reporting of who responded.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A convenient **sample** represents everyone automatically.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

Club membership and volunteering select for interests. Include relevant grade groups and varied reading interests using a planned sampling approach, record who was invited and who responded, and disclose missing groups.

More **response**s from the same selected group do not automatically fix coverage.

Explanations for the additional workbook examples

W1 Define the population

Workbook question · W1

The question is whether all Grade 7 students prefer a longer lunch. Name the **population** and one unsuitable **sample**.

Teacher reasoning · W1

Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

W2 Read the sample result

Workbook question · W2

Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

Teacher reasoning · W2

7/12 of these respondents prefer longer lunch, about 58.3 percent.

Teacher reasoning · W2

The numerator is seven respondents choosing longer lunch and the denominator is all twelve observed respondents. The fraction is 7/12, approximately 58.3 percent.

Say “of these respondents”; the calculation alone supplies no **evidence** about the preferences of students who were not represented.

W3 Repair a claim

Workbook question · W3

A report says all Grade 7 prefers coding after surveying only the coding club. Rewrite it.

Teacher reasoning · W3

Among surveyed coding-club respondents, the observed choices favored coding; the result does not establish the whole grade preference.

W4 Plan independently

Workbook question · W4

Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

Teacher reasoning · W4

Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous **response** records and invited/responded counts.

Explain limits of non**response** and small **sample**s.

W5 Separate invitation and response

Workbook question · W5

Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report non**response**?

Teacher reasoning · W5

No. The 12 nonrespondents may differ. Report 8 of 20 and the limitation.

Graphs and honest scales — W1 Read the scale

A bar is 6 squares long with 2 votes per square. How many votes is that?

12 votes.

One square represents two votes. Six equal squares therefore represent 6 times 2 = 12 votes. Ask the learner to name both the visible length and the unit **scale** before giving the count.

Graphs and honest scales — W2 Compare fairly

Counts are 12 and 15. State the difference and ratio using the true values.

Difference 3; ratio 15/12 = 1.25.

The count difference is 15 - 12 = 3 votes. Taking the larger count relative to the smaller gives 15/12 = 1.25, so 15 is 1.25 times 12, or 25 percent larger. Label which count is the reference for the ratio.

Graphs and honest scales — W3 Repair a misleading graph

Bars for 12 and 15 start at 10. Explain the misleading lengths and repair.

Visible lengths 2 and 5 suggest a 2.5 ratio. Start at zero, use a common **scale** and label counts 12 and 15.

Graphs and honest scales — W4 Draw independently

Draw a **bar chart** for Music 6, Art 9, Coding 3. Include title, category labels, vote units, zero **baseline** and equal **scale**. State two supported comparisons.

Music 6, Art 9, Coding 3 on one **scale**. Art has 3 more than Music and three times Coding. Accept other correct comparisons.

Graphs and honest scales — W5 Limit the claim

> Art has the most votes among 18 **response**s. Does that establish Art is best for learning?

No. Preference counts do not measure learning effectiveness or establish a cause.

Assess independent transfer

> #### Independent question · withhold the key
>
> Selected case: Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

Selected case answer

> #### Teacher key · Selected case answer
>
> Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous **response** records and invited/responded counts.
>
> Explain limits of non**response** and small **sample**s.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Sampling and Representation to Transparent Book Recommender, adapt support and plan a later changed-case check.

Transparent Book Recommender

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 20: Repair Bias in Data. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Use fictional roster cards. Extension: draw several small random **sample**s from a known fictional **population** and compare their proportions.

Independent W4 **evidence**: Use the full grade roster, teacher-managed random selection or a clearly described balanced plan, neutral identical options, anonymous **response** records and invited/responded counts.

Explain limits of non**response** and small **sample**s. Record whether the learner needed no help, a hint or explanation. Fresh retry: plan a **sample** about library opening times for the whole grade.

A library-club-only survey fails coverage; require a full-grade frame and non**response** plan.

Next week: Selection bias and coverage. Reuse this week's explicit contracts and **evidence**.

Readiness and reassessment for Graphs and honest scales

Use graph paper with a marked zero. Extension: show the same values on two clearly labelled **scale**s and discuss visual emphasis.

Independent W4 **evidence**: Music 6, Art 9, Coding 3 on one **scale**. Art has 3 more than Music and three times Coding. Accept other correct comparisons. Record whether the learner needed no help, a hint or explanation.

Fresh retry: Reading 4, Building 8 and Drawing 6; draw all bars from zero with one **scale**. Building has twice Reading, and 2 more than Drawing.

Plan the delayed return

Suggested return points in this level: ai-2/20, ai-2/22, ai-2/25. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent **response** and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Sampling and Representation lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Define a question and target **population**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual **response**s. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Before class, solve the supported practice: Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

Supported practice answer

Reasoned practice answer: 7/12 of these respondents prefer longer lunch, about 58.3 percent.

Check every learner before discussion: Use the worked case for Questions **population**s and **sample**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Selection and **response** are separate. Even a fair invitation **sample** can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Invite only the 15 library-club members instead. Require a **prediction** and a reason before comparing. Expected result: The invitation **process** no longer covers the whole grade through this frame.

Reason: Club members may differ in reading interests; selection is now limited to one group.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

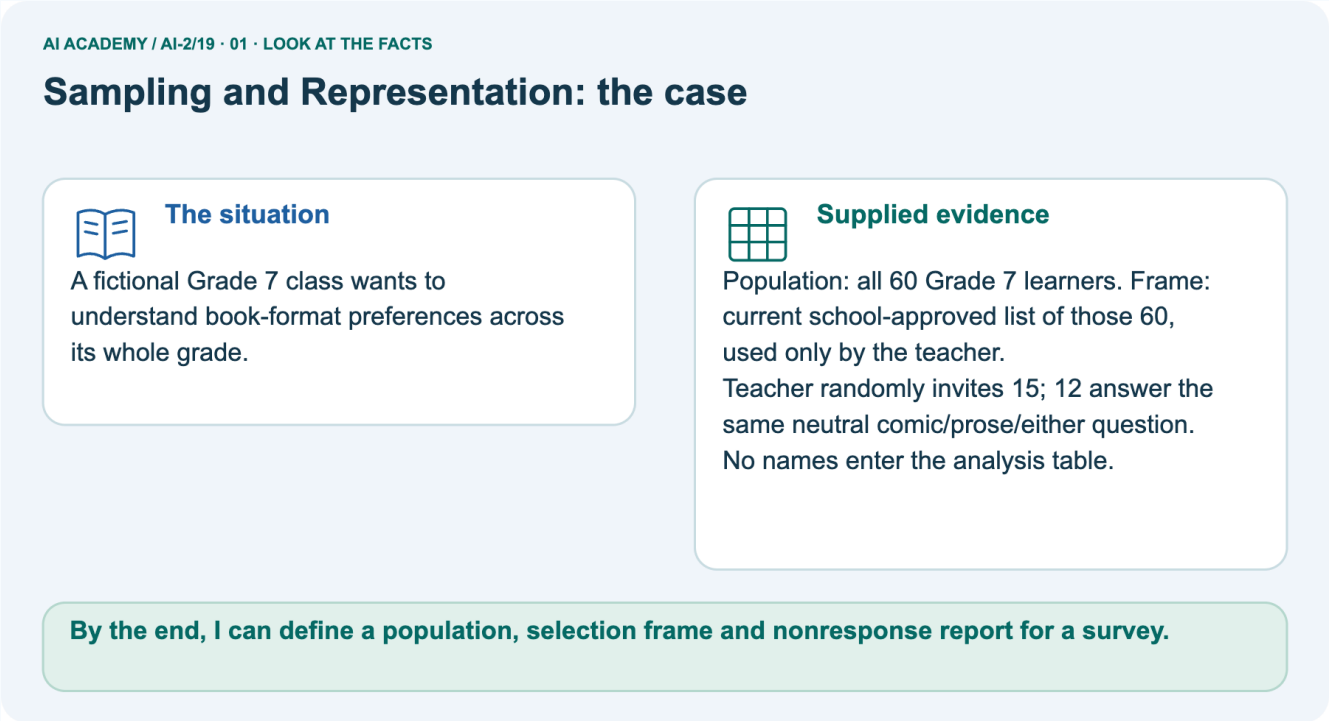
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Sampling and Representation graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sampling and Representation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Grade 7 class wants to understand book-format preferences across its whole grade. Supplied facts: **Population**: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

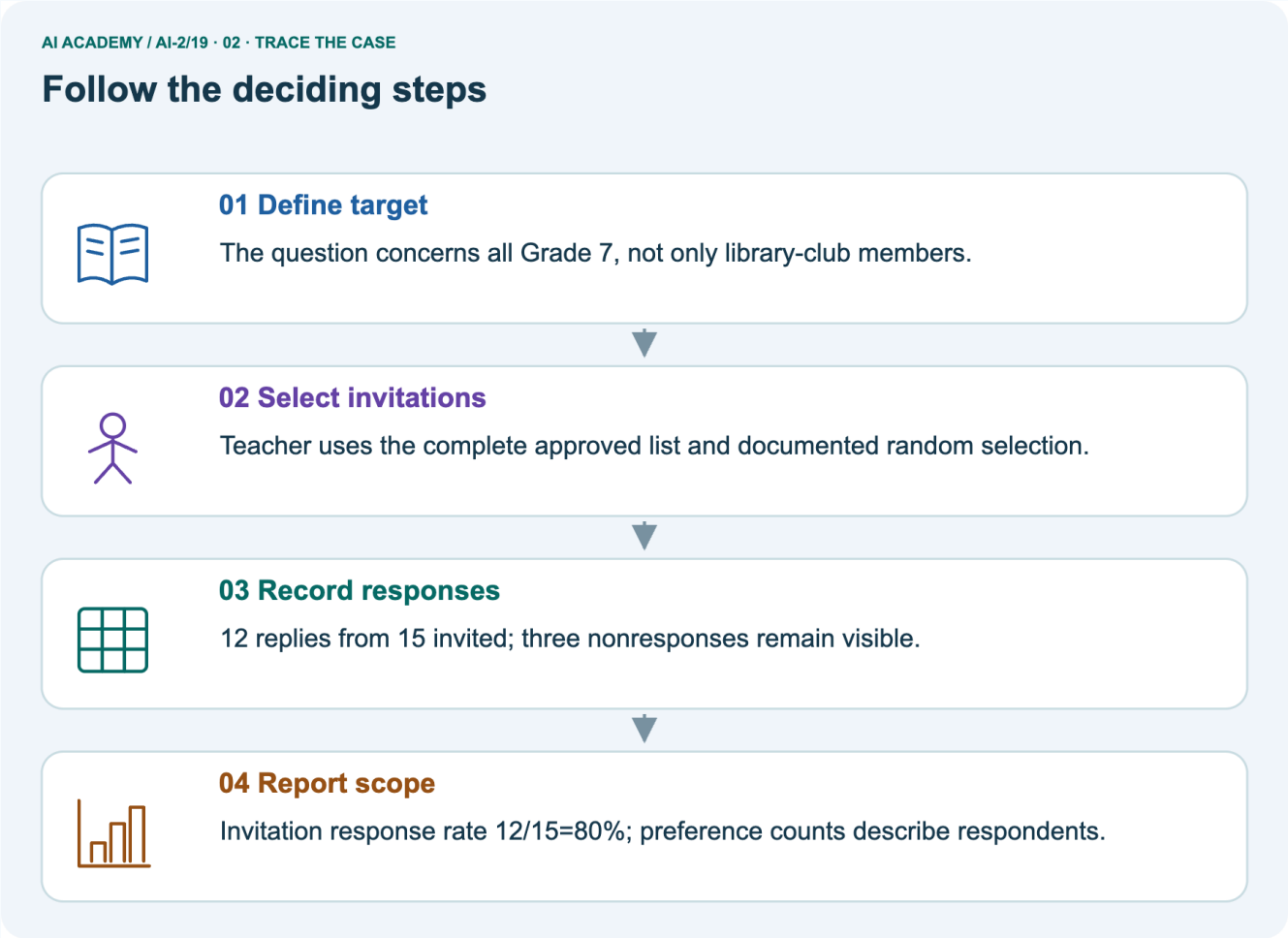
Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

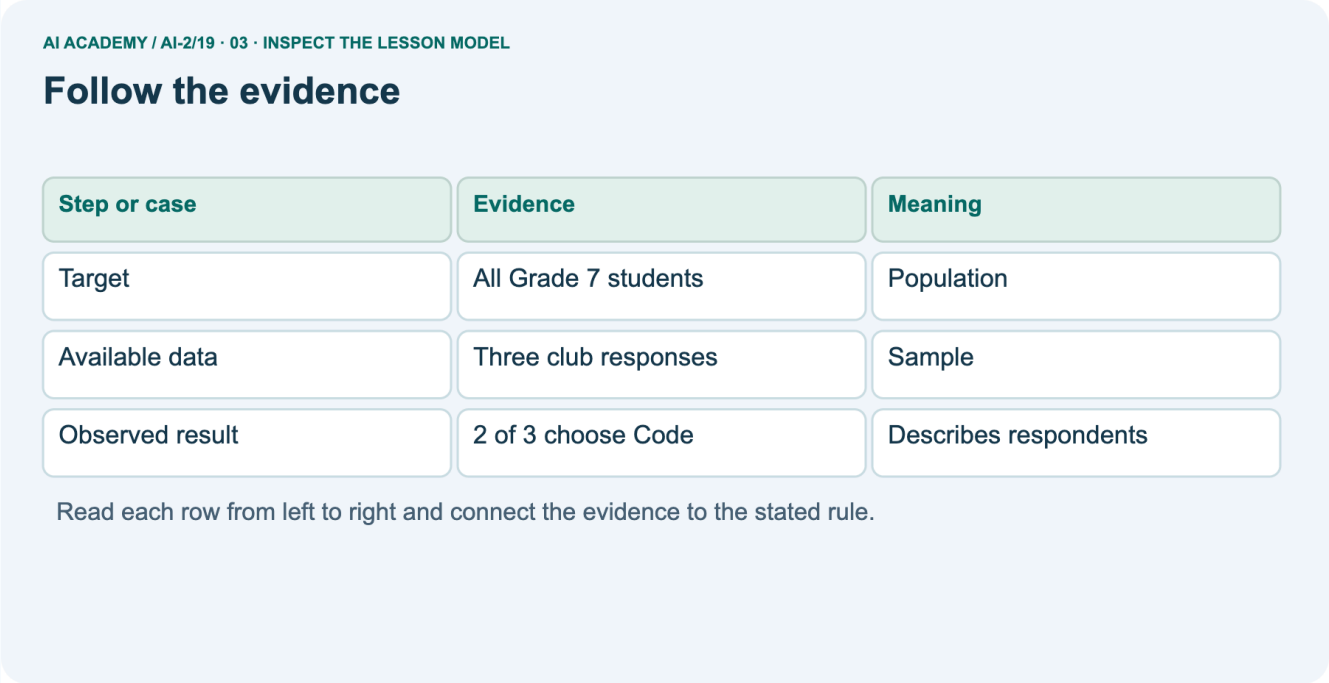
Define target: The question concerns all Grade 7, not only library-club members. → Select invitations: Teacher uses the complete approved list and documented random selection. → Record **response**s: 12 replies from 15 invited; three non**response**s remain visible. → Report scope: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



Follow the labelled picture with the matching explanation. Open the vector diagram

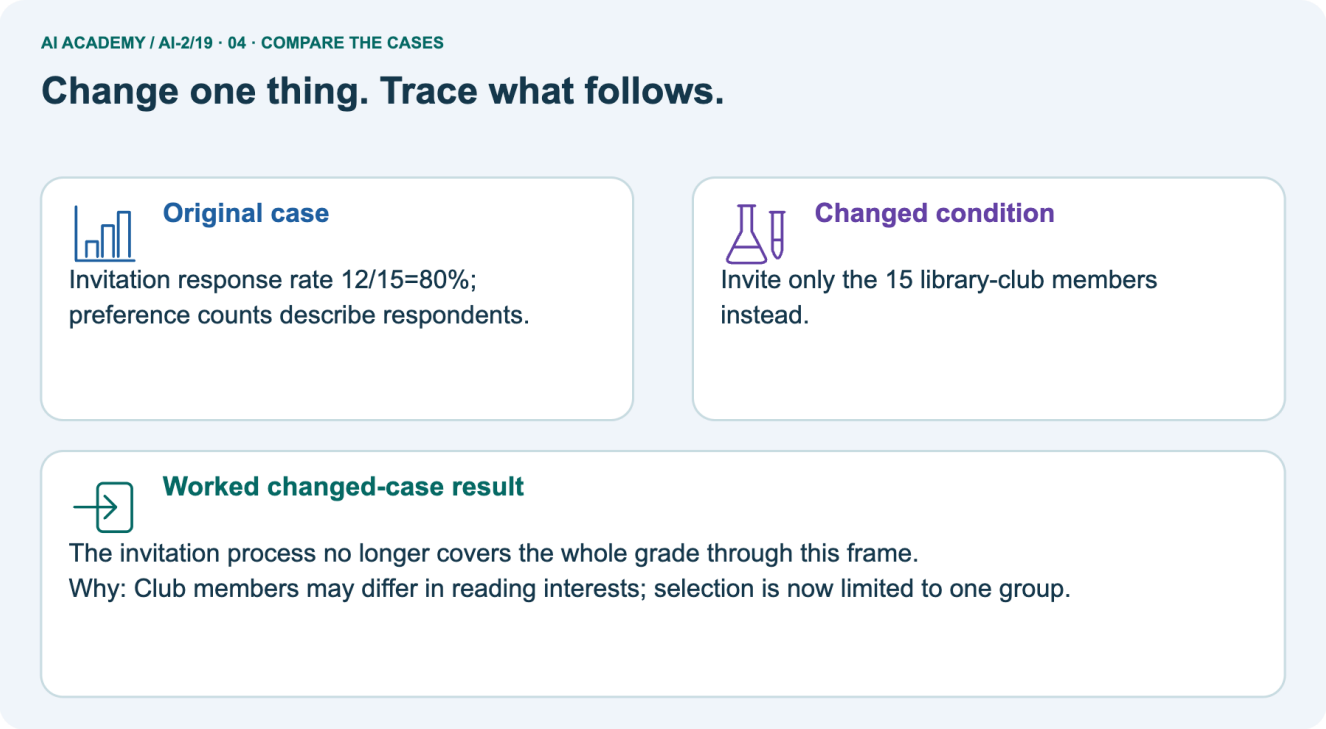
Follow the **evidence**. Target | All Grade 7 students | **Population**. Available data | Three club **response**s | **Sample**. Observed result | 2 of 3 choose Code | Describes respondents

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.
Changed condition: Invite only the 15 library-club members instead.

Expected result: The invitation **process** no longer covers the whole grade through this frame.
Reason: Club members may differ in reading interests; selection is now limited to one group.

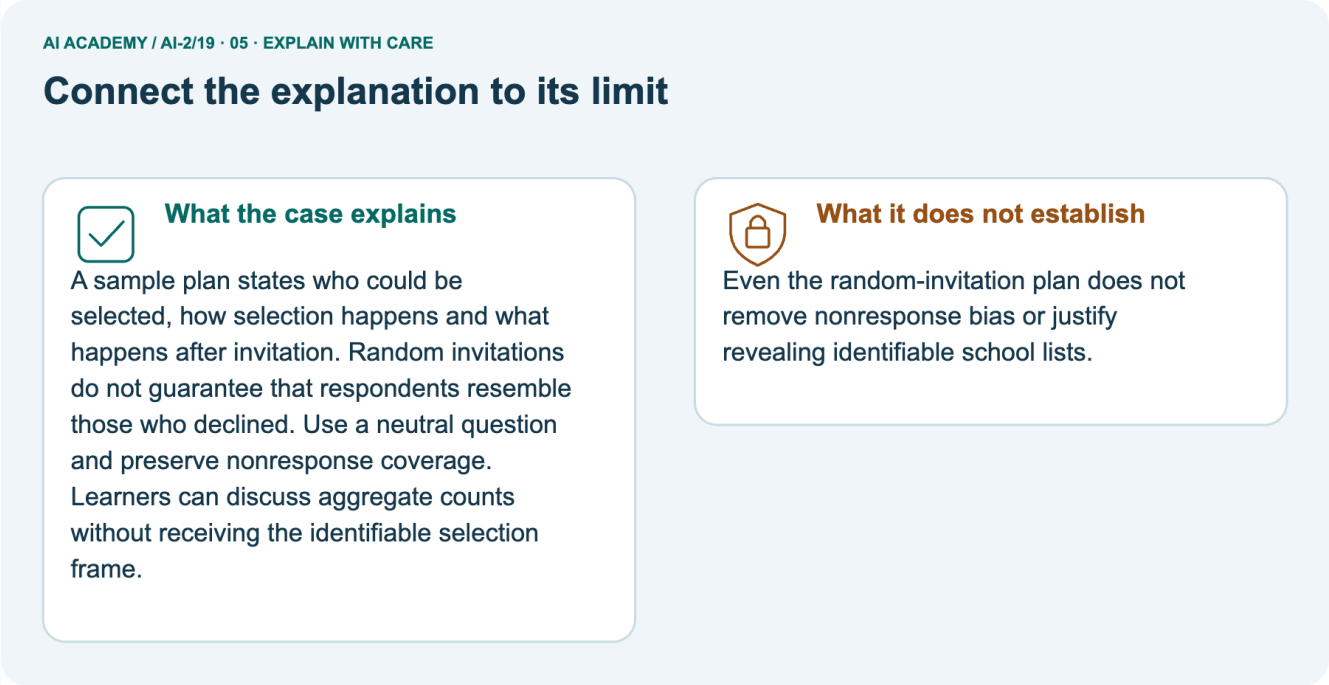
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Club members may differ in reading interests; selection is now limited to one group.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Limit: Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

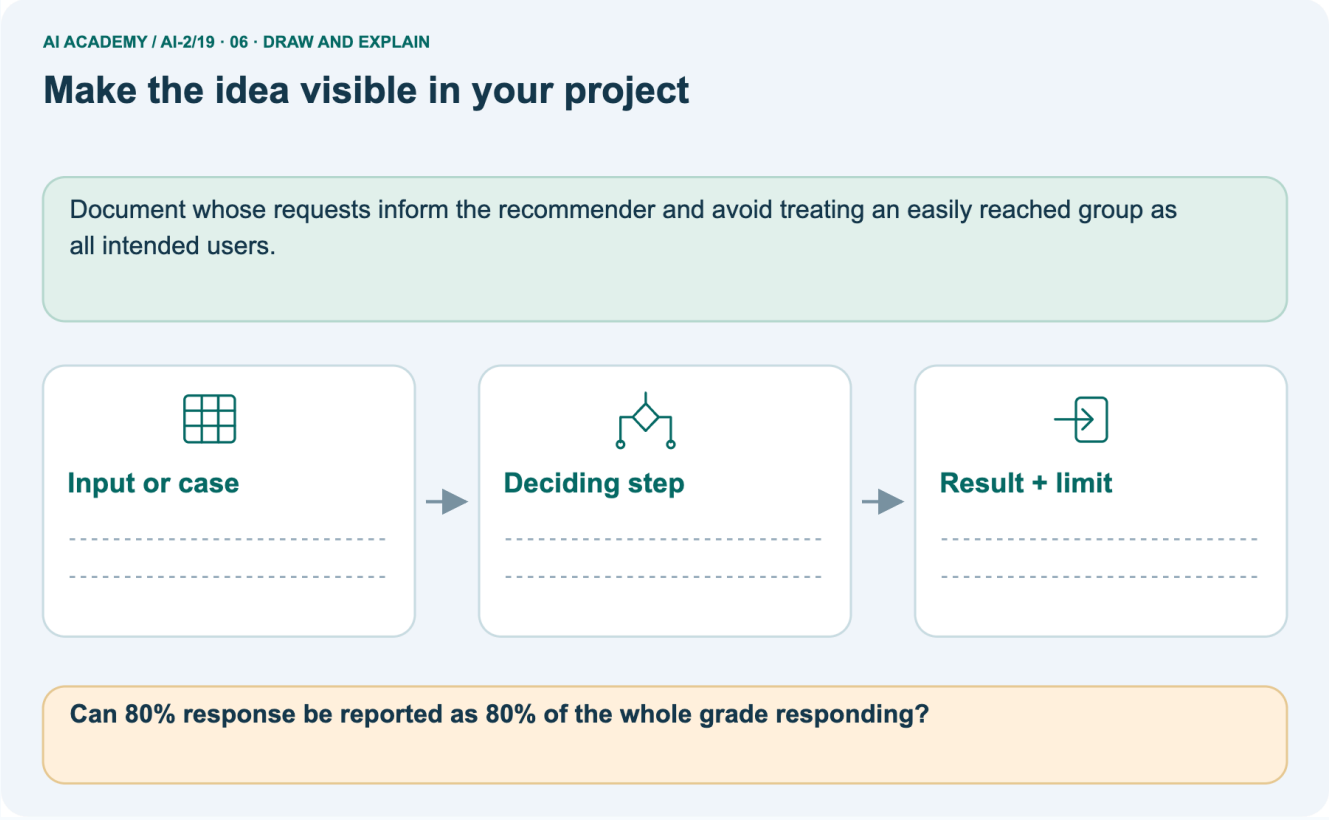
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users. Draw your **input**, deciding step, result and limit.

Supported question: Can **80% response** be reported as **80%** of the whole grade responding?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Twelve of fifteen invited is **80%**; twelve of sixty in the grade is 20%. Denominators answer different questions.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.